

Giant Clam Report

Giant clams are large marine animals that belong to the phylum *Mollusca* and the class *bivalvia* ([National Geographic](#)). Mollusks are soft bodied invertebrates, and bivalves are mollusks with two shells connected by a hinge. Giant clams qualify as mollusks because they have soft internal bodies protected by hard calcium carbonate shells. Like other mollusks, they also have a mantle, gills and a muscular foot during part of their cycle.

I chose giant clams because I thought it was interesting that a clam could grow to be so massive. Before this, I only thought clams were small shellfish found near beaches. Learning that some giant clams can weigh hundreds of pounds made me want to learn more about them.

One of the most famous giant clam species is the *Tridacna gigas*, which is the largest living bivalve in the world ([Britannica](#)). Some giant clams can grow over four feet long and weigh more than 400 pounds. Giant clams live in warm tropical waters in the Indo-Pacific Ocean near coral reefs. They are commonly found around reefs near Australia, the Philippines, and Indonesia. Giant clams need shallow clear water because sunlight is important for their survival.

Giant clams eat in two different ways. First, they are filter feeders, meaning they pull water through their bodies and collect tiny organisms called plankton for food. They also depend on algae called zooxanthellae that live inside their tissues. The algae uses sunlight to create energy through photosynthesis, and the clam receives nutrients from the algae in return. ([NOAA Fisheries](#))

The life cycle of a giant clam starts when eggs and sperm are released into the ocean, where fertilization happens externally. The larvae float in the water for a short time before settling on a reef and growing into adults. Giant clams are hermaphrodites meaning they have both male and female reproductive organs, but they usually release sperm and eggs at different times to reduce self fertilization. ([NOAA Fisheries](#))

Giant clams are not major pearl producers like oysters. While pearls can occasionally form inside them, it is rare and not commercially important. Instead, giant clams are more valuable for their role in coral reef ecosystems, where they help filter water and support marine biodiversity. ([Smithsonian Ocean](#))

Giant clams are considered threatened in some areas due to overharvesting, habitat destruction, and climate change. They have been collected for food, aquarium trade, and shells. Because they grow slowly and live a long time, their populations take a long time to recover. Conservation programs and marine protected areas are helping protect them today. ([Smithsonian Ocean](#)) ([NOAA Fisheries](#))

One interesting fact is that giant clams can live for over 100 years. Their mantles can also display bright colors like blue, green, purple and gold because of the algae living inside them. Another common myth is that they can trap humans, but this is false because their shells close too slowly to be dangerous. ([Smithsonian Ocean](#))

I watched a video about giant clams that explained how they survive in coral reef environments, it showed how they rely heavily on sunlight due to their symbiotic algae and how they filter water for nutrients. The video also shows how giant clams slowly react to movement by closing their shells and explaining their importance in keeping reef

ecosystems healthy., it helped show that giant clams are much more important to life than they first appear. ([Giant Clams](#))